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MTH633 QUIZ NO 1 FILE

SOLVED BY SANAULLAH KHAN

CONTACT NO:03066793776

Subject Enrolment

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 - GDB's
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Final project (MTH620)

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MTH633 - Group Theory (Quiz NO. 1) Quiz Start Time: 04:34 PM, 11 December 2022

Question # 1 of 10 (Start time: 04:34:24 PM, 11 December 2022)

Total Marks: 1

If $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 1 & 4 \end{pmatrix}$ is a permutation then, $\alpha(3) = 1$.

Select the correct option

Reload Math Equations

True

☒

False

☐

Click to Save Answer & Move to Next Question

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MTH633 - Group Theory (Quiz NO. 1) Quiz Start Time: 04:34 PM, 11 December 2022

Question # 2 of 10 (Start time: 04:34:36 PM, 11 December 2022)

Total Marks: 1

An orbit of a permutation p is an equivalence class under the relation :
 $a \sim b \Leftrightarrow b = p^n(a)$, for some n in $\mathbb{Z}$.

Select the correct option

Reload Math Equations

True

☒

False

☐

Click to Save Answer & Move to Next Question

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MTH633 - Group Theory (Quiz NO. 1) Quiz Start Time: 04:34 PM, 11 December 2022

Question # 3 of 10 (Start time: 04:35:30 PM, 11 December 2022)

Total Marks: 1

Let G be a group and $a \in G$. Then the subgroup $H = \{ a^n | n \in \mathbb{Z} \}$ of G is called the cyclic subgroup of G generated by a , and is denoted by $\langle a \rangle$.

Select the correct option

Reload Math Equations

True

☒

False

☐

Click to Save Answer & Move to Next Question

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MTH633 - Group Theory (Quiz NO. 1) Quiz Start Time: 04:34 PM, 11 December 2022

Question # 4 of 10 (Start time: 04:35:40 PM, 11 December 2022)

Total Marks: 1

A permutation of a set B is a function from B to B that is both one to one and into.

Select the correct option

Reload Math Equations

False

☒

True

☐

Click to Save Answer & Move to Next Question

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MTH633 - Group Theory (Quiz NO. 1) Quiz Start Time: 04:34 PM, 11 December 2022

Question # 5 of 10 (Start time: 04:35:49 PM, 11 December 2022)

Total Marks: 1

Z_{12} is generated by $\{2, 4, 6, 8, 10\}$.

Select the correct option

Reload Math Equations

True

☐

False

☒

Click to Save Answer & Move to Next Question

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MTH633 - Group Theory (Quiz NO. 1) Quiz Start Time: 04:34 PM, 11 December 2022

Question # 6 of 10 (Start time: 04:36:00 PM, 11 December 2022)

Total Marks: 1

A mapping $\phi: G \rightarrow G'$ is called isomorphic if it is

Select the correct option

Reload Math Equations

ϕ is surjective and homomorphic

☐

ϕ is bijective and homomorphic

☒

ϕ is injective and homomorphic

☐

ϕ is bijective and linear

☐

Click to Save Answer & Move to Next Question

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MTH633 - Group Theory (Quiz NO. 1) Quiz Start Time: 04:34 PM, 11 December 2022

Question # 7 of 10 (Start time: 04:36:11 PM, 11 December 2022)

Total Marks: 1

An orbit of a permutation p is an equivalence class under the relation :
 $a \sim b \Leftrightarrow b \neq p^n(a)$, for some n in $\mathbb{Z}$.

Select the correct option

Reload Math Equations

☒	False
☐	True

Click to Save Answer & Move to Next Question

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MTH633 - Group Theory (Quiz NO. 1) Quiz Start Time: 04:34 PM, 11 December 2022

Question # 8 of 10 (Start time: 04:36:21 PM, 11 December 2022)

Total Marks: 1

G is cyclic group if $G = \langle a \rangle$ for some a in G .
If $|a| = n$, $a^i = a^j$ iff ---

Select the correct option

Reload Math Equations

☐	$n i \times j$
☒	$n i - j$
☐	$n i + j$
☐	$n j - i$

Click to Save Answer & Move to Next Question

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MTH633 - Group Theory (Quiz NO. 1) Quiz Start Time: 04:34 PM, 11 December 2022

Question # 9 of 10 (Start time: 04:36:51 PM, 11 December 2022)

Total Marks: 1

A cycle of length --- is called a transposition.

Select the correct option

Reload Math Equations

☐	1
☒	2
☐	4
☐	3

Click to Save Answer & Move to Next Question

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MTH633 - Group Theory (Quiz NO. 1) Quiz Start Time: 04:34 PM, 11 December 2022

Question # 10 of 10 (Start time: 04:37:01 PM, 11 December 2022)

Total Marks: 1

Is $\langle \bar{5} \rangle$ the generator of Z_6 ?

Select the correct option

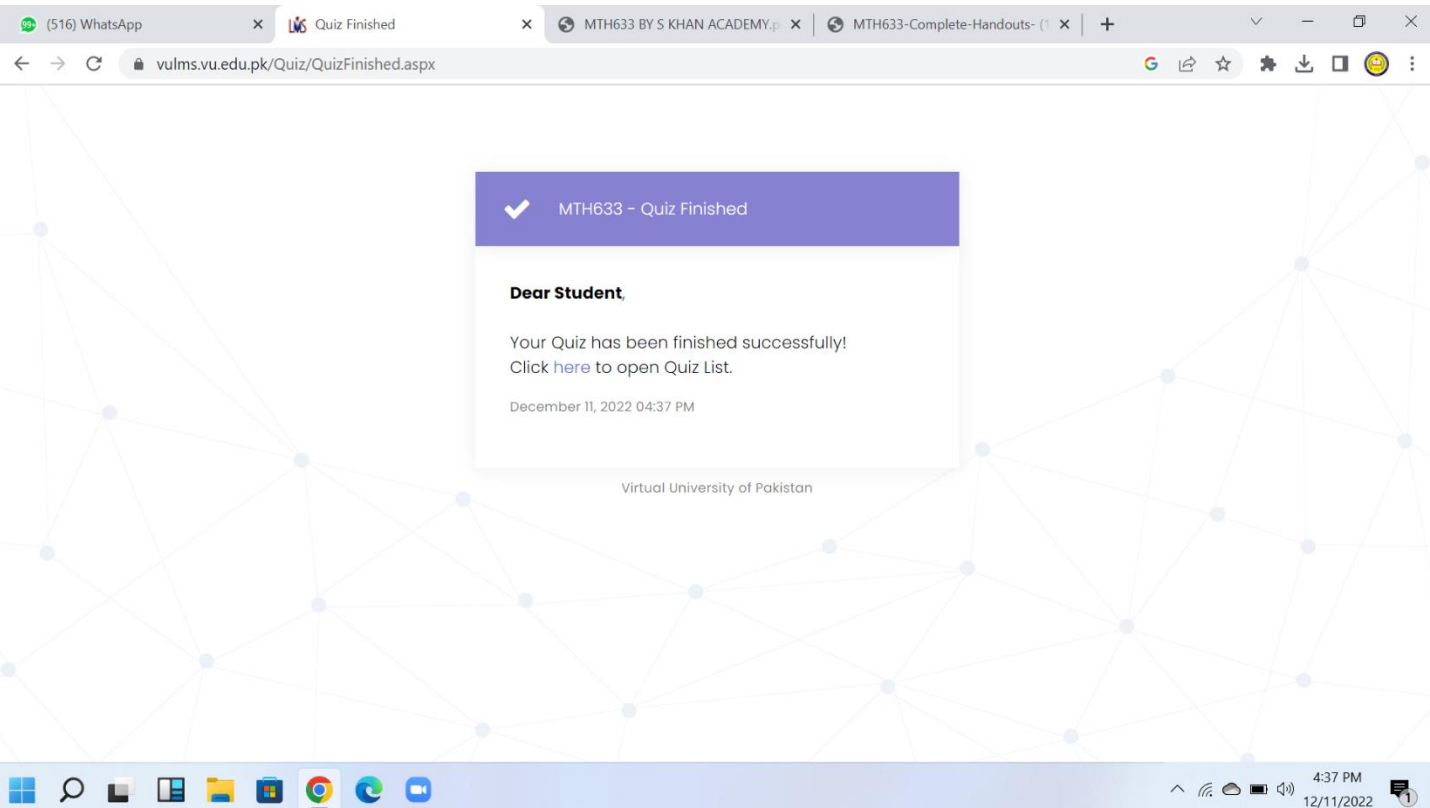
Reload Math Equations

☒	Yes
☐	No

Click to Save Answer & Move to Next Question

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MTH633 - Group Theory (Quiz NO. 1) Quiz Start Time: 05:38 PM, 11 December 2022

Question # 1 of 10 (Start time: 05:38:16 PM, 11 December 2022)

Total Marks: 1

Z_{18} is generated by $\{1, 5, 7, 11, 13, 17\}$.

Select the correct option

Reload Math Equations

True



False



Click to Save Answer & Move to Next Question

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MTH633 - Group Theory (Quiz NO. 1) Quiz Start Time: 05:38 PM, 11 December 2022

Question # 2 of 10 (Start time: 05:38:28 PM, 11 December 2022)

Total Marks: 1

Z_6 is cyclic.

Select the correct option

Reload Math Equations

False



True



Click to Save Answer & Move to Next Question

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MTH633 - Group Theory (Quiz NO. 1) Quiz Start Time: 05:38 PM, 11 December 2022

Question # 3 of 10 (Start time: 05:38:36 PM, 11 December 2022)

Total Marks: 1

Cayley's theorem states that every group of permutations is isomorphic to a group.

Select the correct option

Reload Math Equations

True

☒

False

☐

Click to Save Answer & Move to Next Question

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MTH633 - Group Theory (Quiz NO. 1) Quiz Start Time: 05:38 PM, 11 December 2022

Question # 4 of 10 (Start time: 05:39:32 PM, 11 December 2022)

Total Marks: 1

An orbit of a permutation p is an equivalence class under the relation :
$$a \sim b \Leftrightarrow b = p^n(a), \text{ for some } n \text{ in } \mathbb{Z}.$$

Select the correct option

Reload Math Equations

False

☐

True

☒

Click to Save Answer & Move to Next Question

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MTH633 - Group Theory (Quiz NO. 1) Quiz Start Time: 05:38 PM, 11 December 2022

Question # 5 of 10 (Start time: 05:39:43 PM, 11 December 2022)

Total Marks: 1

Is $\langle \bar{5} \rangle$ the generator of Z_6 ?

Select the correct option

Reload Math Equations

☐	No
☒	Yes

Click to Save Answer & Move to Next Question

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MTH633 - Group Theory (Quiz NO. 1) Quiz Start Time: 05:38 PM, 11 December 2022

Question # 6 of 10 (Start time: 05:39:51 PM, 11 December 2022)

Total Marks: 1

G is cyclic group if $G = \langle a \rangle$ for some $a \in G$.
If $|a| = n$, $a^i = a^j$ iff — —

Select the correct option

Reload Math Equations

☐	$n i \times j$
☐	$n j - i$
☒	$n i - j$
☐	$n i + j$

Click to Save Answer & Move to Next Question

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MTH633 - Group Theory (Quiz NO. 1) Quiz Start Time: 05:38 PM, 11 December 2022

Question # 7 of 10 (Start time: 05:40:01 PM, 11 December 2022)

Total Marks: 1

The group generated by a contains elements of the form - - -

Select the correct option

Reload Math Equations

- ☐ $\langle a \rangle = \{a, a^2, a^3, \dots, a^{n+1}, e\}$
- ☒ $\langle a \rangle = \{a, a^2, a^3, \dots, a^{n-1}, e\}$
- ☐ $\langle a \rangle = \{a, a^2, a^3, \dots, a^n, e\}$
- ☐ $\langle a \rangle = \{a, a^2, a^3, \dots, a^{n-1}\}$

Click to Save Answer & Move to Next Question

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MTH633 - Group Theory (Quiz NO. 1) Quiz Start Time: 05:38 PM, 11 December 2022

Question # 8 of 10 (Start time: 05:40:11 PM, 11 December 2022)

Total Marks: 1

A cycle of length --- is called a transposition.

Select the correct option

Reload Math Equations

- ☒ 2
- ☐ 1
- ☐ 3
- ☐ 4

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MTH633 - Group Theory (Quiz NO. 1) Quiz Start Time: 05:38 PM, 11 December 2022

Question # 9 of 10 (Start time: 05:40:18 PM, 11 December 2022)

Total Marks: 1

For each positive divisor k of n , the group $\langle a \rangle$ has exactly one subgroup of order k , namely $\langle a^{n/k} \rangle$.

Select the correct option

Reload Math Equations

☐	False
☒	True

Click to Save Answer & Move to Next Question

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MTH633 - Group Theory (Quiz NO. 1) Quiz Start Time: 05:38 PM, 11 December 2022

Question # 10 of 10 (Start time: 05:40:28 PM, 11 December 2022)

Total Marks: 1

A mapping $\phi: G \rightarrow G'$ is called isomorphic if it is

Select the correct option

Reload Math Equations

☐	ϕ is surjective and homomorphic
☐	ϕ is injective and homomorphic
☐	ϕ is bijective and linear
☒	ϕ is bijective and homomorphic

Click to Save Answer & Move to Next Question

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